Lattice Gas Automata

a historical overview

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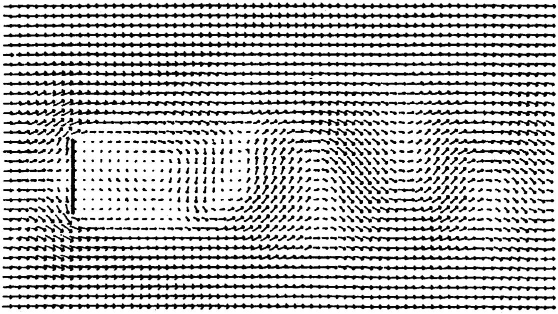
Outline

Bénard-von Kàrmànn street



Wake behind a plate at $Re = 70$ (D. d'Humières and P. Lallemand, 1985)

Discovery in Flow Dynamics May Aid Car, Plane Design

By Philip J. Hilts
Washington Post Staff Writer

Scientists at the Los Alamos National Laboratory and in France have developed a new computerized method to predict the flow of air and fluids that may improve the aerodynamics of cars and planes, track weather systems more reliably and expedite development of "Star Wars" weapons.

The new method of predicting flow dynamics, still in the early testing stage, could allow relatively unsophisticated computers to make calculations that can now be done only by new-generation supercomputers over days or weeks.

According to the method's developers, this has led researchers and defense specialists to question whether the discovery may not be of great value to the Soviet Union, which lags behind the United States in computer sophistication and capacity.

Brosl Hasslacher of the Los Alamos laboratory, who created the new computer model in collaboration with Uriel Frisch of the Observatoire d'Nice in France, said Defense Department officials at the lab questioned whether his technical paper, expected to be published in the next few weeks in Physical Review Letters, should be classified to keep it out of Soviet hands.

But, he said, everyone quickly decided that the basic mathematics are in an area of research well-

See FLOW, A10, Col. 1

A new paradigm

A virtual simplified micro-world is constructed as an automaton universe based not on a realistic description of interacting particles (as in molecular dynamics simulations) but merely on the conservation laws of microscopic physics and the laws of symmetry of macroscopic physics of fluids.

Frisch, Haslacher, Pomeau, 1986 ; Wolfram, 1986

And Feynman ...

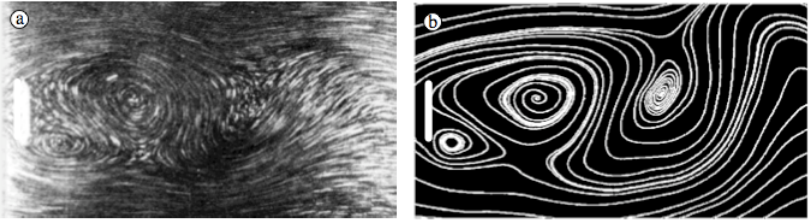
... told us to explain it like this :

We have noticed in nature that the behavior of a fluid depends very little on the nature of the individual particles in that fluid. For example, the flow of sand is very similar to the flow of water or the flow of a pile of ball bearings (*). *We have therefore taken advantage of this fact to invent a type of imaginary particle that is especially simple for us to simulate. This particle is a perfect ball bearing that can move at a single speed in one of six directions. The flow of these particles on a large enough scale is very similar to the flow of natural fluids.*

Physics Today (February, 1989)

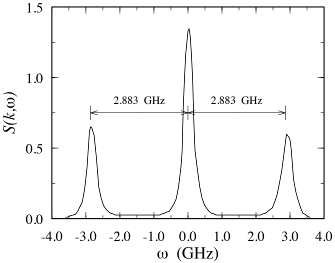
(*) *Reynolds similarity law*

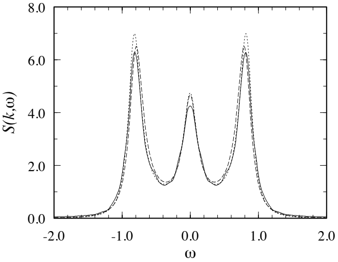
Hydrodynamics $\rightarrow$ von Kàrmàn vortex street



Streamlines at $Re = 200$: (a) L. Prandtl, 1934 ; (b) LGA, J.P. Rivet, 1987

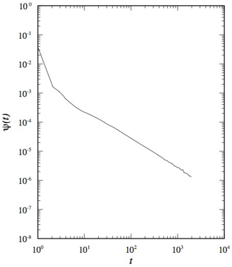
LGA : reservoir of excitations $\longrightarrow$ Hydrodynamic fluctuations : $S(k,\omega)$





Liquid Ar, Fleury & Boon, 1969. LGA, Grosfils, Boon & Lallemand, 1992.

Long time behavior of the velocity autocorrelation function



$$\psi(t) \simeq \frac{D-1}{D} \frac{v_0}{b} \frac{1-d}{d} (4\pi(\nu + D_s)t)^{-D/2} \sim t^{-1} \text{ for } D = 2$$

mode coupling theory and LGA, van der Hoef, Frenkel, Ernst, 1990.

Basics of LGA

The lattice gas automaton

Set of particles moving in a space-time discretized universe :

- a regular D-dimensional lattice $\mathcal{L}$
- at discrete time steps, $t = m\Delta t$ ($m \in \mathrm{N}$)
- $\mathcal{L}$: V nodes labeled by D-dimensional position vectors $\mathbf{r} \in \mathcal{L}$
- each node has b channels ($i, j, \ldots = [0, b-1]$ or $[1, b]$)

At given time t, a channel can be empty or occupied :
occupation variable $\quad \overline{n}_i^*(\mathbf{r}, t) = 0 \quad$ or $\quad n_i^*(\mathbf{r}, t) = 1$
If channel i at node $\mathbf{r}$ is occupied, there is *one and only one* particle at node $\mathbf{r}$ with velocity $\mathbf{c}_i \Longrightarrow$ **Exclusion principle.**
The set of velocities is such that $\mathbf{r} + \mathbf{c}_i\Delta t \in \mathcal{L}$.

LGA dynamics

Propagation phase (deterministic and reversible) :
particles move according to $\mathbf{c}_i$:
$n_i^*(\mathbf{r} + \mathbf{c_i}\Delta t, t + \Delta t) = n_i^*(\mathbf{r}, t)$

Collision phase (local) :
particles are redistributed according to collision rules
$\Longrightarrow n_i^*(\mathbf{r} + \mathbf{c_i}\Delta t, t + \Delta t) = n_i^*(\mathbf{r}, t) + \Omega_i^*(\{n_i^*\})$

Conservation laws (collisional invariants)

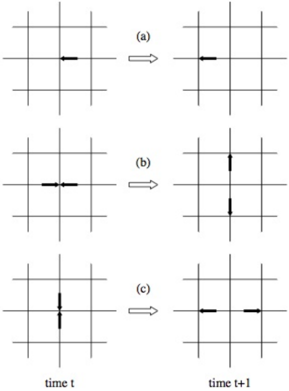
Mass : $\sum_i n_i^*(\mathbf{r}, t)_{BC} = \sum_i n_i^*(\mathbf{r}, t)_{AC}$

Momentum : $\sum_i \mathbf{c}_i \, n_i^*(\mathbf{r}, t)_{BC} = \sum_i \mathbf{c}_i \, n_i^*(\mathbf{r}, t)_{AC}$

Energy : $\sum_i |\mathbf{c}_i|^2 n_i^*(\mathbf{r}, t)_{BC} = \sum_i |\mathbf{c}_i|^2 n_i^*(\mathbf{r}, t)_{AC}$

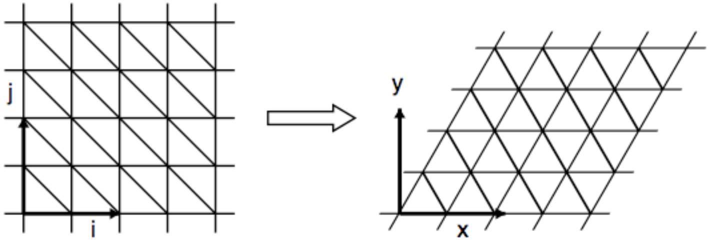
Energy is trivially conserved in single speed LGA (e.g. $|\mathbf{c}_i| = 1$)
$\Longrightarrow$ non-thermal LGA.

2-D square lattice : HPP (*)



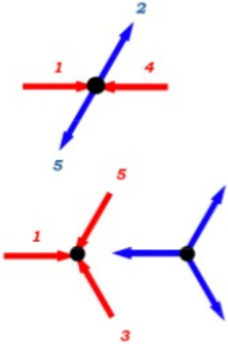
(*) Hardy, de Pazzis, Pomeau, 1973

2-D square lattice gas : (i) spurious invariant (ii) non-isotropic hydrodynamics $\Longrightarrow$ 2-D triangular lattice (*)



(*) Frisch, Haslacher, Pomeau, 1986

2-D triangular lattice : FHP (*)



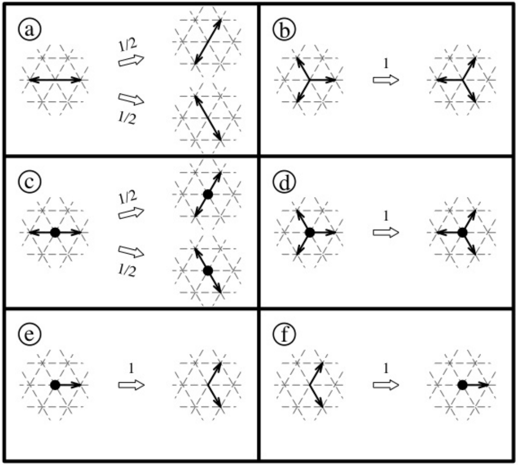
$$\sigma^{(2)}(n_1 n_4 \hat{n}_2 \hat{n}_5 - n_2 n_5 \hat{n}_1 \hat{n}_4)$$

$$\sigma^{(3)}(n_1 n_3 n_5 \hat{n}_2 \hat{n}_4 \hat{n}_6 - n_2 n_4 n_6 \hat{n}_1 \hat{n}_3 \hat{n}_5)$$

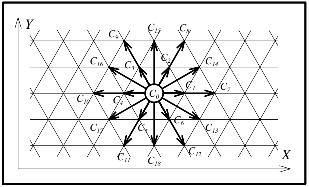
Basic 2- and 3-particle collisions

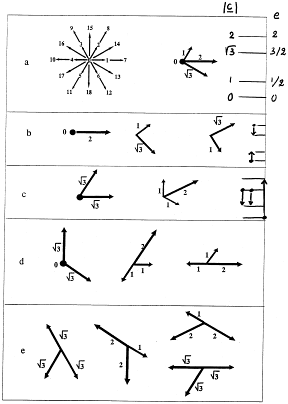
(*) Frisch, Haslacher, Pomeau, 1986

FHP-2 Collisions



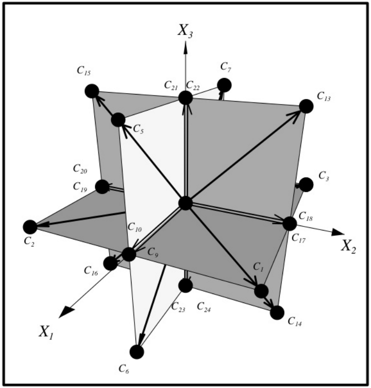
Thermal LGA - GBL (*) model





(*) Grosfils, Boon, Lallemand, 1992

3-D LGA : FCHC (*) : 3-D projection of the $F4$ lattice



(*) d'Humières, Lallemand, Frisch, 1986

From LGA to the Lattice Boltzmann Equation

Microdynamics ($\sim$ Newton's equations for LGA)

State of a node (Boolean field)

$\mathrm{n}(\mathbf{r}, t) = \{n_i^*(\mathbf{r}, t) \quad ; \quad \mathbf{r} \in \mathcal{L} \quad ; \quad i = 0, ..., b-1\}$

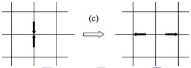
$n_i^*(\mathbf{r}, t)$: occupation number of channel i at node $\mathbf{r}$ at time t

2 step updating rule : Collision $+$ Propagation

$\Longrightarrow$ **Microdynamical Equation**

$$\boxed{n_i^*(\mathbf{r} + \mathbf{c}_i, t+1) = n_i^*(\mathbf{r}, t) \; + \; \Omega_i^*(\{n_j^*(t)\})}$$

Ex : $\Omega_i^* = \overline{n}_i^* \, n_{i+1}^* \, \overline{n}_{i+2}^* \, n_{i+3}^* \; - \; n_i^* \, \overline{n}_{i+1}^* \, n_{i+2}^* \, \overline{n}_{i+3}^*$



Probabilistic automaton : $\Omega_i^* = \sum_{s,s'} (s_i' - s_i)\, \xi_{s,s'}^* \prod_j n_j^{*\,s_j}\, \overline{n}_j^{*\,s_j}$;

$\xi_{s,s'}^*$: state $s \rightarrow s'$ transition selection Boolean variable

Probabilistic automaton : $\Omega_i^* = \sum_{s,s'} (s_i' - s_i)\, \xi_{s,s'}^* \prod_j n_j^{*\,s_j}\, \overline{n}_j^{*\,s_j}$;

$\xi_{s,s'}^*$: state $s \to s'$ transition selection Boolean variable

Mean field dynamics

$$\langle n_i^*(\mathbf{r} + \mathbf{c}_i, t+1) \rangle = \langle n_i^*(\mathbf{r}, t) \rangle + \langle \Omega_i^*(\{ n_j^*(\mathbf{r}, t) \}) \rangle$$

- Def : $f_i(\mathbf{r}, t) = \langle n_i^*(\mathbf{r}, t) \rangle_{ne}$
- Boltzmann factorization “ansatz” : $\langle \Omega_i^*(n_j^*) \rangle_{ne} = \Omega_i(f_j)$
  Correlations before collision are neglected ($\sim$ Molecular chaos)
- Transition probabilities : $A_{s,s'} = \langle \xi_{s,s'}^* \rangle_{ne}$

$\Longrightarrow$ **Semi-detailed balance** : $\sum_s A_{s \to s'} = \sum_s A_{s' \to s}$

Microdynamics : $n_i^*(\mathbf{r}+\mathbf{c}_i, t+1) = n_i^*(\mathbf{r}, t) \; + \; \Omega_i^*(\{n_j^*(t)\})$

$\Downarrow$

Mean field : $\langle n_i^*(\mathbf{r}+\mathbf{c}_i, t+1)\rangle = \langle n_i^*(\mathbf{r}, t)\rangle \; + \; \langle\Omega_i^*(\{n_j^*(\mathbf{r}, t)\})\rangle$

$\Downarrow$

Def : $f_i(\mathbf{r}, t) \; = \; \langle n_i^*(\mathbf{r}, t)\rangle_{ne}$; Factorization : $\langle\Omega_i^*(\, n_j^* \,)\rangle_{ne} \; = \; \Omega_i(\, f_j \,)$

$\Longrightarrow$ **Lattice Boltzmann Equation** $\qquad (\sim \partial_t f_i + \mathbf{c}_i \cdot \partial_{\mathbf{r}} f_i = Coll\,)$

$$\boxed{f_i(\mathbf{r}+\mathbf{c}_i, t+1) \; - \; f_i(\mathbf{r}, t) \; = \; \Omega_i\,(f)}$$

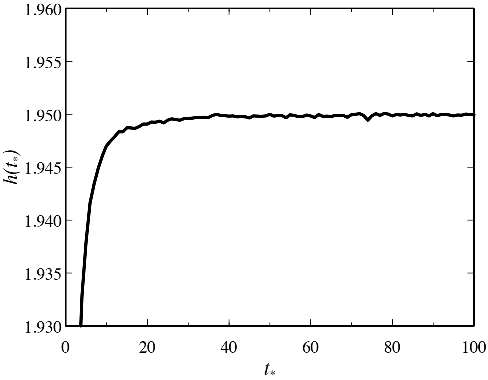
Collision term : $\Omega_i \; = \; \sum_{s,s'} (s_i' - s_i)\, A_{s,s'} \, \prod_j \, f_j^{s_j} \, (1 - f_j)^{1-s_j}$

$\Longrightarrow$ **H theorem** $\Longrightarrow f_i^{\mathrm{eq}} \; = \; [1 \; + \; \exp(-\alpha \; + \; \frac{1}{2}\,\beta\,|\mathbf{c}_i|^2)]^{-1}$

$\sim$ Fermi-Dirac distribution

$\alpha \sim$ chemical potential ; $\beta \sim$ reciprocal Temperature

H-Theorem $\longrightarrow$ Entropy



FHP-1 : $h(t_*) = -\sum_i \left[f_i(t_*) \ln f_i(t_*) + \overline{f}_i(t_*) \ln \overline{f}_i(t_*) \right]$ (Tribel, Boon, 1997)

Macrodynamics and Thermodynamics

Macrodynamics

Space and time behavior of slow variables

Density : $\rho(\mathbf{r},t) = \sum_i f_i(\mathbf{r},t) = \sum_i \langle n_i^*(\mathbf{r},t)\rangle$ (per channel : $d = \rho/b$)

Current density : $\mathbf{j}(\mathbf{r},t) = \sum_i \mathbf{c}_i f_i(\mathbf{r},t)$ (mean velocity $\mathbf{u} = \mathbf{j}/\rho$)

Energy density : $\varepsilon(\mathbf{r},t) = \frac{1}{2}\sum_i |\mathbf{c}_i|^2 f_i(\mathbf{r},t)$ $(+\sum_i U_i)$

Long-wavelength ($k\ell \to 0$), long-time ($\omega\tau \to 0$) dynamics

- microscopic characteristic scales :

ℓ : lattice unit length ; $\tau = \frac{\ell}{c}$: lattice time scale ; ($c = |\mathbf{c}|$)

- Macroscopic characteristic scales :

Macroscopic length scale : $\Lambda \gg \ell$ with $\frac{\ell}{\Lambda} = \epsilon \ll 1$

Macroscopic time scales :
(i) Propagation : $t_s = \frac{1}{c_s\, k} = \frac{\Lambda}{c_s} \simeq \frac{\Lambda}{c} = \frac{\Lambda}{\ell}\frac{\ell}{c} = \epsilon^{-1}\,\tau$
(ii) Dissipation : $t_D = \frac{1}{\nu\, k^2} = \frac{\Lambda^2}{\nu} \simeq \frac{\Lambda^2}{\ell c_s} = \frac{\Lambda^2}{\ell^2}\frac{\ell}{c} = \epsilon^{-2}\,\tau$

$\Longrightarrow$ multiscale expansion $\Longrightarrow$ Macrodynamics

Hydrodynamic equations

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0$$
$$\partial_t (\rho \mathbf{u}) + \nabla \cdot (g(\rho) \rho \mathbf{u}\mathbf{u}) = -\nabla p + \nabla \cdot \mathbb{S}$$

- Pressure : $p = \frac{c^2}{D} \rho (1 - \mathcal{O}(u^2/c^2))$
  When $|\mathbf{u}| \ll c \Rightarrow$ compressibility equation : $\frac{1}{\rho} \frac{\partial \rho}{\partial p} = \frac{D}{\rho c^2}$;
  sound velocity : $c_s = \frac{c}{\sqrt{D}}$
- Viscous stress tensor $\mathbb{S}$
  Symmetry : $S_{\alpha\beta} = \nu_s \partial_\alpha (\rho u_\beta) + (\frac{D-2}{D} \nu_s + \nu_B) \partial_\alpha (\rho u_\alpha)$
- ν_s =kinematic viscosity ; $\nu_B =$ bulk viscosity.
- non-Gallilean invariance factor : $g(\rho) = \frac{D}{D+2} \frac{1-2d}{1-d}$

Incompressible fluid

$\frac{d\rho}{dt} = 0 \longrightarrow \nabla \cdot \mathbf{u} = 0$

Scaling :

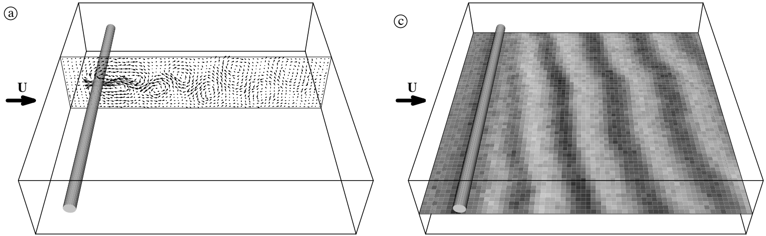
$\tilde{\mathbf{r}} = \frac{1}{L_H}\mathbf{r}\,; \qquad \tilde{t} = \frac{g(\rho_0)u_H}{L_H}t = \frac{t}{T_H}\,; \qquad \tilde{\mathbf{u}} = \frac{1}{u_H}\mathbf{u}\,; \qquad \tilde{p} = \frac{1}{\rho_0\, g(\rho_0)\, u_H^2}p$

$Re = \frac{u_H L_H}{\nu} \longrightarrow \tilde{R} = \frac{u_H L_H}{\nu(\rho_0)/g(\rho_0)} = R^\star\, L_H\, M_a = \frac{M_a}{K_n^\star}$

$$\begin{aligned} \nabla \cdot \tilde{\mathbf{u}} &= 0 \\ \partial_{\tilde{t}}\, \tilde{\mathbf{u}} + \tilde{\mathbf{u}} \cdot \nabla \tilde{\mathbf{u}} &= -\nabla\, \tilde{p} + \frac{1}{\tilde{R}} \Delta \tilde{\mathbf{u}} \end{aligned}$$

Turbulent flow $\longrightarrow$ Limitations : ν, L_H

3-D von Kàrmàn street



Three-dimensional incompressible flow around a cylinder in wind tunnel
$Re = 1.57\, R_c$ (a) 74 T_H ; (c) 112 T_H (FCHC, Rivet, 1993)

Hydrodynamic fluctuations

LGA : (i) Large number of degrees of freedom
(ii) Boolean microscopic nature + Stochastic dynamics
$\Longrightarrow$ LGA : Reservoir of excitations ($\sim$ real fluid)

$\delta f_i(\mathbf{r}, t) = f_i(\mathbf{r}, t) - f^{eq} \quad \Longrightarrow$ Linearized Boltzmann Equation

$$\delta f_i(\mathbf{r} + \mathbf{c}_i, t + 1) \; = \; \delta f_i(\mathbf{r}, t) \; + \; \sum_j \Omega_{ij} \, \delta f_j(\mathbf{r}, t)$$

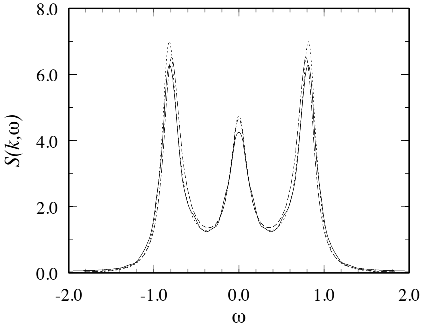
$$\Longrightarrow \delta \tilde{f}_i(k, s) \; = \; \sum_j \frac{e^{s+\imath k \, c_j}}{e^{s+\imath k \, c_j} - \mathbf{1} - \Omega_{ij}} \, \delta f_j(k, t = 0)$$

Resolvent $\Longrightarrow$ Eigen vectors and values $\Longrightarrow$ LGA modes

$\Longrightarrow$ Dynamic structure factor

$$S(k,\omega) = 2\,\mathcal{R}e \sum_{i,j} \langle \delta\tilde{f}_i(k, s = \imath\omega)\delta f_j(k, t = 0)\rangle + S^0(k)$$

$$S^0(k) = \frac{\sum_j f_j^{eq}(1-f_j^{eq})}{\sum_j f_j^{eq}} \text{ (static structure factor)}$$



Thermal LGA (19 velocities)

Statistical Thermodynamics

Equation of state : $p = \beta^{-1}\,\rho\,(1 + B_2\,\rho + ...)$
ρ = density per node ; $B_2 = \frac{1}{2}\,\rho^{-2}\,\sum_i f_i^2$; compressibility : $\chi = \frac{1}{\rho}\frac{\partial\rho}{\partial p}$

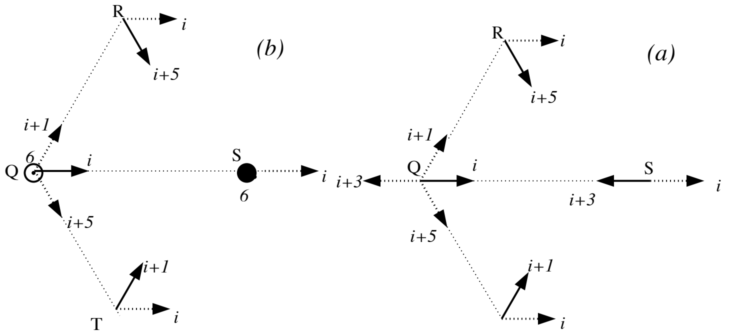
Density fluctuation correlations :
Def : $\delta f_i(k,t) = \sum_{\mathbf{r}} e^{-\imath\mathbf{k}\cdot\mathbf{r}}\,[f_i(\mathbf{r},\mathbf{c}_i;t) - f]$

Static structure factor : $\rho\,S(k) = \frac{1}{T}\sum_{t=1}^{T}\sum_{i,j}\delta f_i(k,t)\delta f_j(k,t)$

LGA (with local collisions) : $\rho\,S^0(k) = \sum_j f_j^{eq}(1 - f_j^{eq}) \equiv \sum_j \kappa_j^{(2)}$

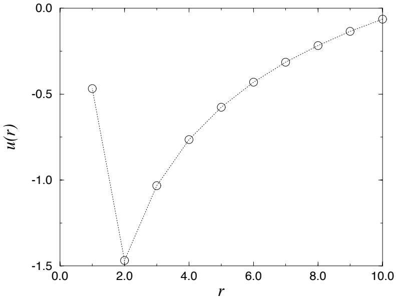
$\longrightarrow S^0(k) = \rho\,\beta^{-1}\,\chi^0$ (continuous systems : $\lim_{k\to 0} S(k) = \rho\,\beta^{-1}\,\chi^0$)

LGA with non-local collisions (*) (Long distance momentum transfer)



(*) Appert, Zaleski, 1993 ; Tribel, Boon, 1995

$\longrightarrow$ "pair potential" ($\sim$ rate of momentum transfer)
$u(r) = \pm\gamma f^2 (1-f)^2 F(r)$ with distribution $F(r)$($e.g. \sim r^{-z}$)



$\longrightarrow S(k) = \frac{\rho\,\beta^{-1}\,\chi^0}{1-\gamma\langle r\rangle\kappa^{(3)}}$; $\kappa^{(3)} \equiv f(1-f)(1-2f)$; $\gamma\langle r\rangle\sim$ NLI interactions

Static structure factor : $\frac{S(k)}{S^0(k)} = 1 + f\, h(k)$; $\quad h(k) = \mathcal{F}\{(g(r) - 1)\}$

$g(r) = \exp[-\beta\phi(r)]$: pair correlation function ; $\phi(r)$: potential of mean force

Static structure factor : $\frac{S(k)}{S^0(k)} = 1 + f\, h(k)$; $h(k) = \mathcal{F}\{(g(r) - 1)\}$

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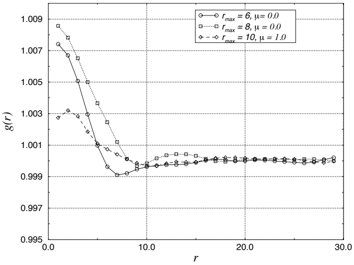
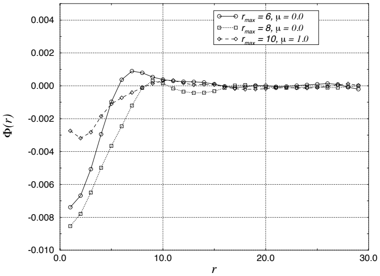
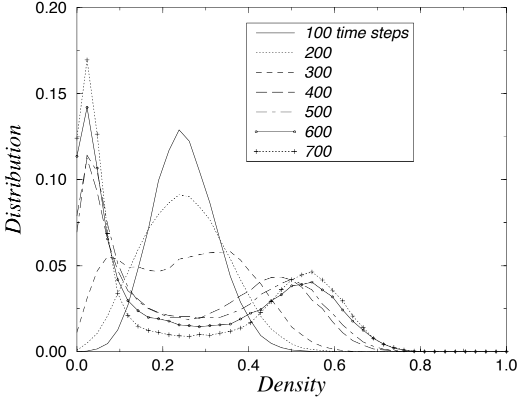


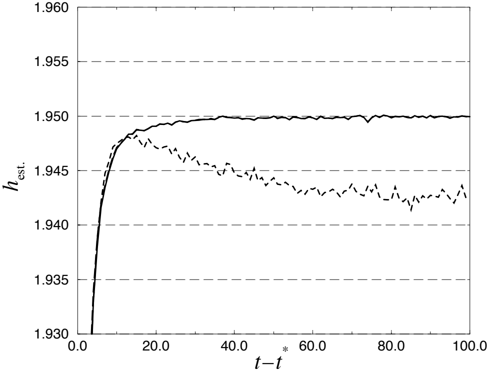


$$c_s^2 = \frac{\partial p}{\partial \rho} = c_0^2 \left(1 - \gamma \langle r \rangle \kappa^{(3)}\right) \longrightarrow \langle r \rangle \nearrow \longrightarrow c_s^2 < 0$$

$\Longrightarrow$ Acoustic modes become unstable $\Longrightarrow$ Spinodal decomposition



Violation of detailed balance $\longrightarrow$ Entropy

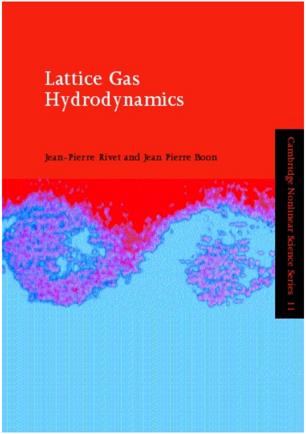


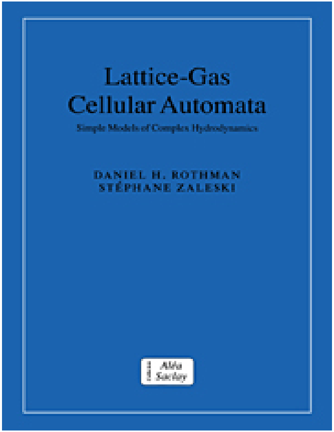
$$h(t-t_*) = -\sum_i \left[f_i(t-t_*)\,\ln f_i(t-t_*) + \bar{f}_i(t-t_*)\,\ln \bar{f}_i(t-t_*) \right]$$ (Tribel, Boon, 1997)

Beyond basics

- Multiphase Fluids
- Multispecies Fluids
- Reactive (R-D) systems
- Turbulent diffusion
- LGA with dynamical geometry
- . . . .
  - Lattice Boltzmann Equation method $\longrightarrow$ Applications

Bibliography





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